Hw1 report

Part 1: Data Loading and First Look

Q1:

The data has 1250 rows of patients , for each patient there are 25 columns of data.

Q2:

the value_counts of the conversations_per_day feature:

```
print(dataset.conversations_per_day.value_counts())
```

```
conversations_per_day
3     218
2     204
5     179
4     168
1     108
6     107
7      94
8      54
9      42
10     29
11     16
13      8
12      7
14      6
16      5
15      3
17      1
29      1
Name: count, dtype: int64
```

This feature describes the number of conversations a patient had in a day.

It is considered Ordinal because the number of conversations has a logical, natural sequence ordering to it, and it groups the num of conversations into separate categories.

Q3:

features description

Feature	Brief Explanation	Type
patient_id	The ID of the patient. In our data in a Range of: [1,1250]	Categorical
age	The age of the patient. In a Range of: [3,84]	Continuous
sex	The sex of the patient. Male or Female	Categorical
weight	The weight of the patient. In a Range of: [5, 114.9]	Continuous
blood_type	The Blood Type of the patient.	Categorical
current_location	The GPS Coordinates of the patient, Latitude and longitude	Ordinal
num_of_siblings	Number of Siblings a Patient has.	Ordinal
happiness_score	Score of how happy the patient is.	Ordinal
household_income	Household income of the patient.	Continuous
conversations_per_day	How many Conversations a per has a day	Ordinal
sugar_levels	Sugar blood levels of a patient	Continuous
sport_activity	how active the patient is in sports.	Ordinal
pcr_date	When he had the PCR tests done	Categorical
PCR_01 … to	10 PCR tests done to test for COVID on the same patient, Negative values could indicate that the patient tested negative, Positive ones could mean that he tested positive for COVID,	Other

… PCR10	could be used to take the majority of tests to determine if he is sick.	

Q4:

It is important because as we mentioned, we should never use test sets to make decisions about which algorithms to use or for improving or tuning algorithms. So, we always want out two sets to be disjoint and never train on test or test on train set, using the same seed prevents this leakage and ensures reproducibility.

Part 2: Missing Values

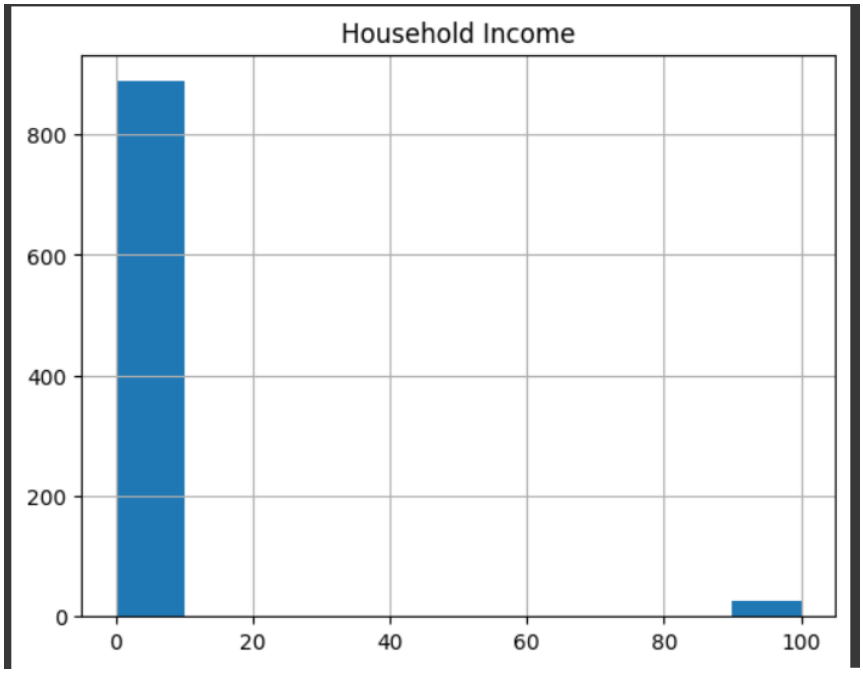
Q5:

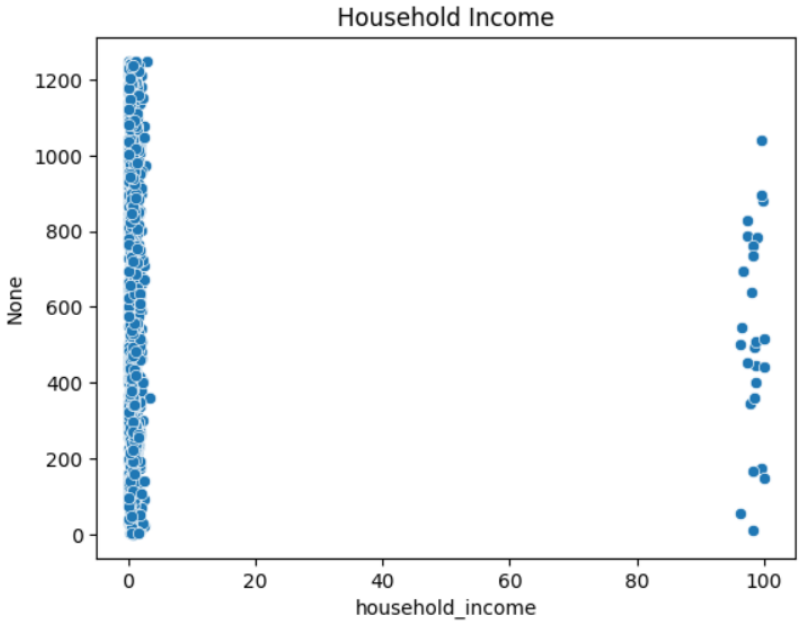
Features with missing values:

Training set	test set
patient_id 0 age 0 sex 0 weight 0 blood_type 0 current_location 0 num_of_siblings 0 happiness_score 0 household_income 86 conversations_per_day 0 sugar_levels 0 sport_activity 0 pcr_date 0 PCR_01 0 PCR_02 0 PCR_03 0 PCR_04 0 PCR_05 0 PCR_06 0 PCR_07 0 PCR_08 0 PCR_09 0 PCR_10 0 spread 0 risk 0 dtype: int64	test.isnull().sum() patient_id 0 age 0 sex 0 weight 0 blood_type 0 current_location 0 num_of_siblings 0 happiness_score 0 household_income 23 conversations_per_day 0 sugar_levels 0 sport_activity 0 pcr_date 0 PCR_01 0 PCR_02 0 PCR_03 0 PCR_04 0 PCR_05 0 PCR_06 0 PCR_07 0 PCR_08 0 PCR_09 0 PCR_10 0 spread 0 risk 0 dtype: int64
86 null values in the household_income feature, all else are non-null.	23 null values in the household_income feature, all else are non-null.

Q6:

We made a histogram and a scatterplot for convenience:





As we can see, there are outliers that have a household_income value of that is approximately 100 (around 90-100), while most of the rest have an income of around the range 0-10

Q7:

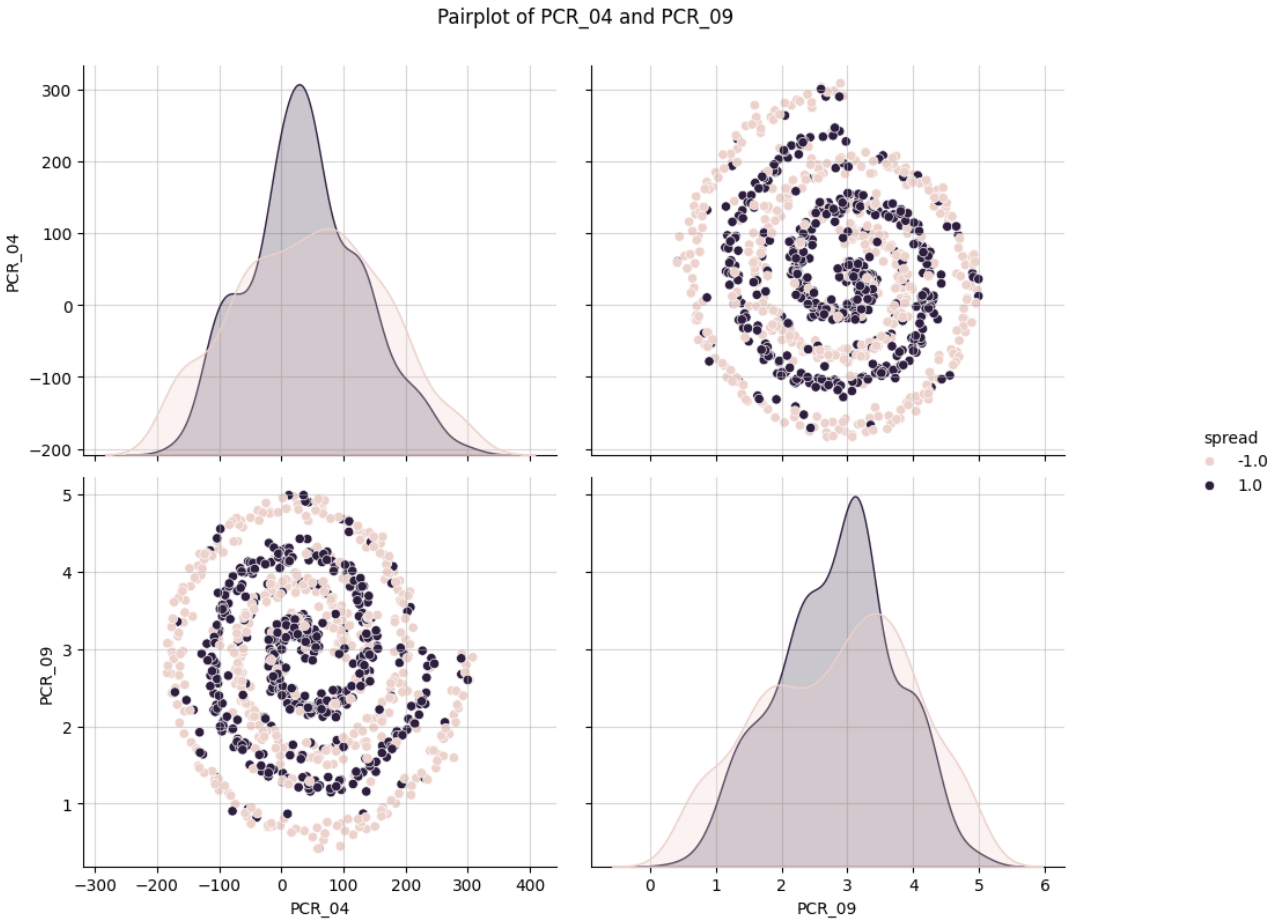
The mean of the training set is: 3.5278993435448576

The median of the training set is: 0.7

The difference between them is a result of there being outliers in the data, and we know that Mean is more affected by outliers since they skew the sum, Therefore, the median is more robust and is the preferred filling method, because it represents the data better.

Part 3: Warming up with k-Nearest Neighbors

Q8:



As we can see from the pair plot of 4-9, we can see that there is a **better separability in the pair4-9** than other available pairs , thus it is useful for predicting the spread.

Q9:

Time complixity of K-NN:

Distances: computation of dist between one point and the test point takes O(d), then, for all points with test point O(md)

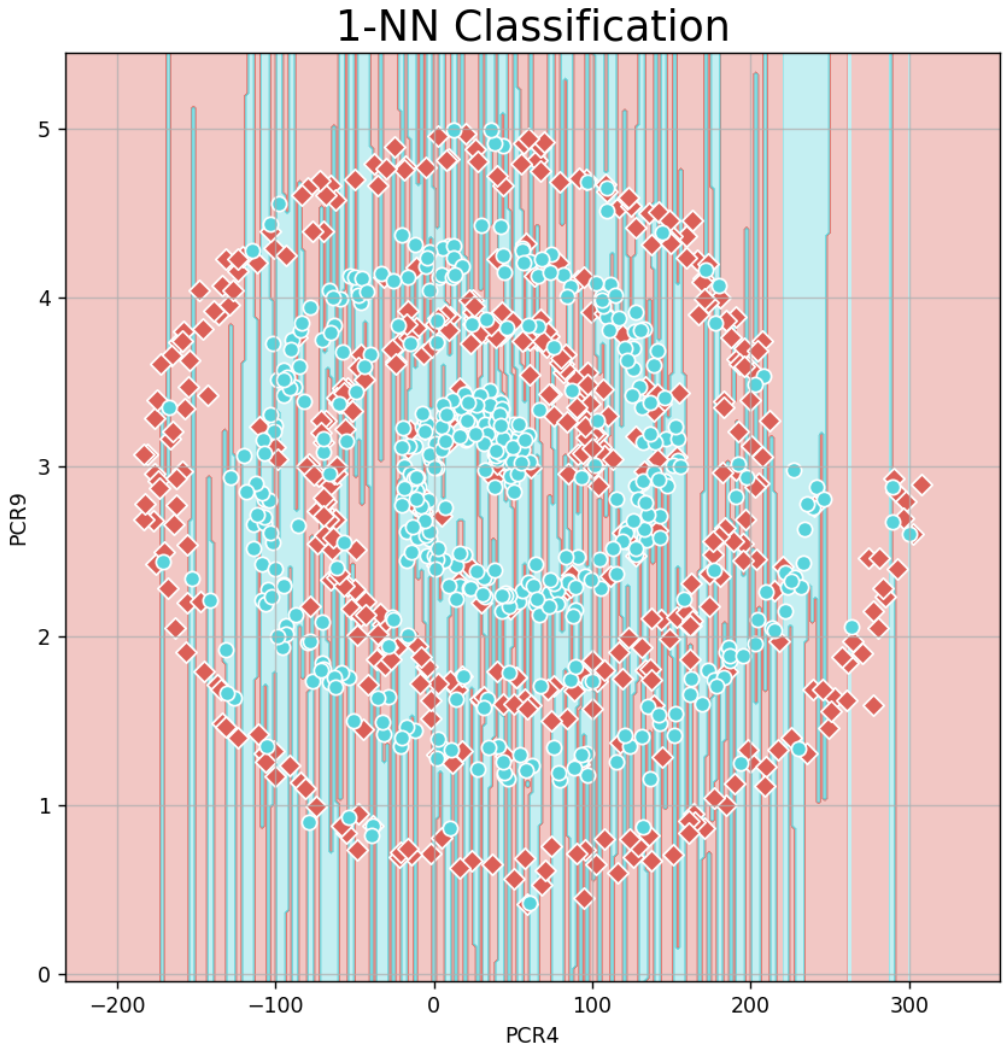
Indices: argpartition takes O(m)

Labels: from indices we get the labels directly in O(k)

Majority: to find the majority in k labels we need O(k)

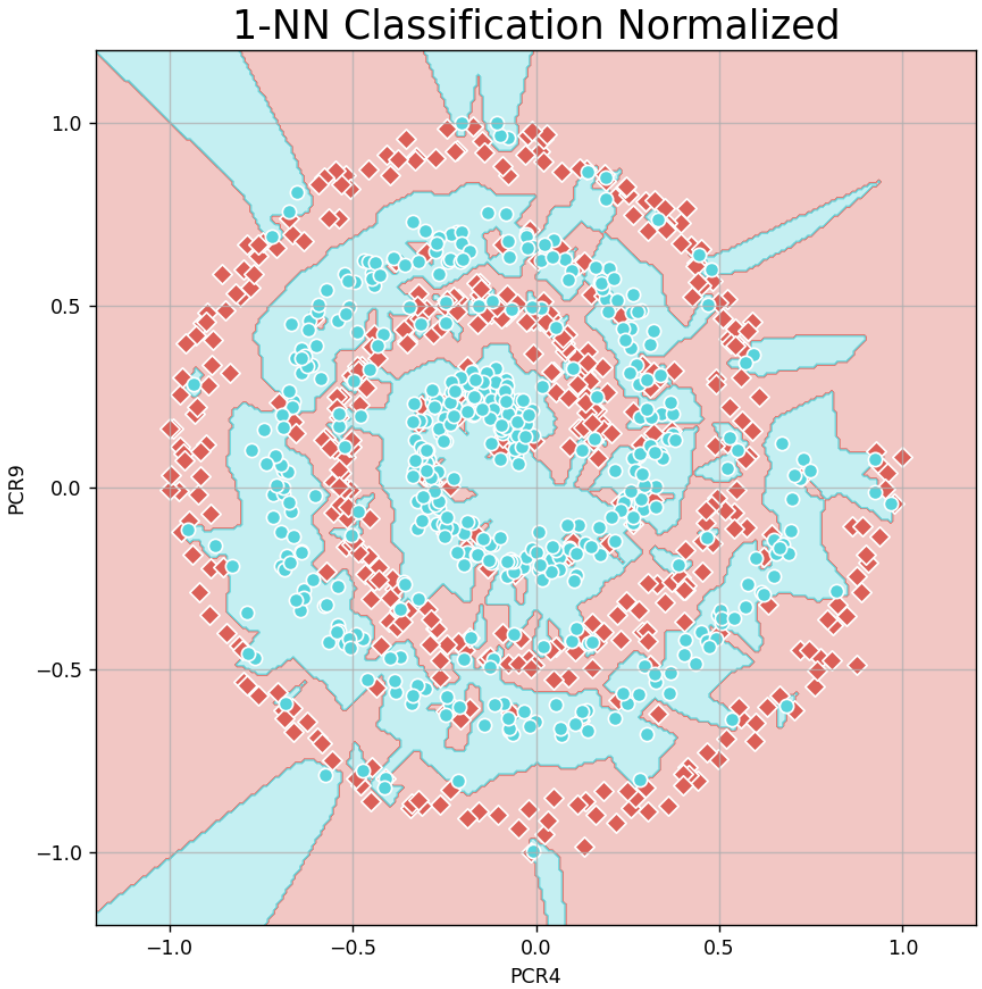
Then, the final complexity is O(md)

Q10:




```
Training score is: 100.0%
Testing score is: 58.8%
```

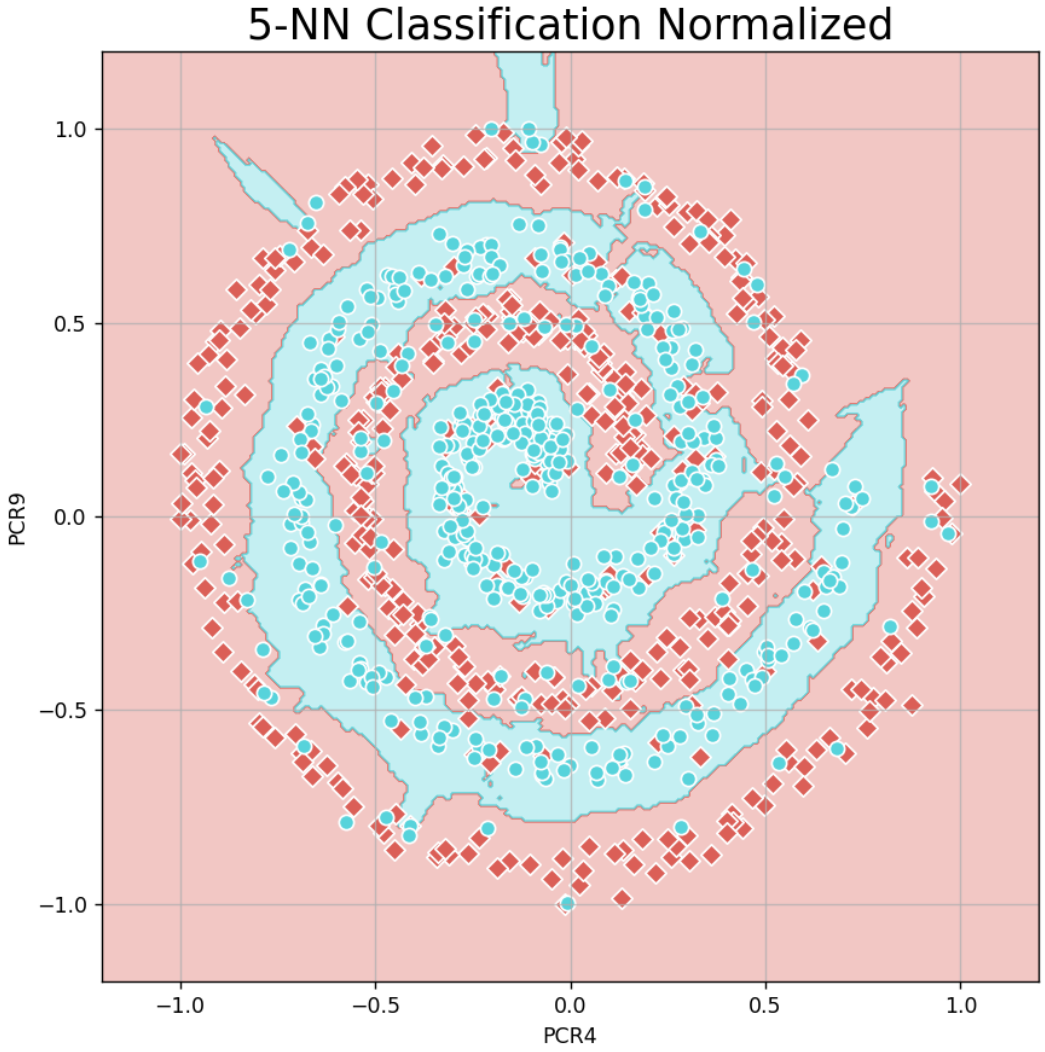
Q11:




```
Training score is: 100.0%
Testing score is: 75.6%
```

The reason why we got better result with normalization is because different features can have different ranges, So if one feature ahas a significantly bigger range, it will be stronger (more dominant) in the distance which leads to biased inaccurate results

Q12:




```
Training score is: 84.3%
Testing score is: 85.2%
```

5-NN has **lower training score** because by using 5nn we are taking the majority vote of 5 nearest labels which will always be less accurate than taking the **actual** label

5-NN has a **Higher test score** because 5nn reduces overfitting to the train data, thus it is less sensitive to outliers in the training data which leads to higher accuracy.

5-NN has more **defined** rigid **decision boundaries** in contrast to 1NN which is very fragmented and is very sensitive to outliers.

Q13:

The first feature is uniformly distributed from 2 to 5 , therefore, there is no 'bias' to certain values in the distrubtion, so normalizing it into the range -1,1 should be uniform and has good variability.

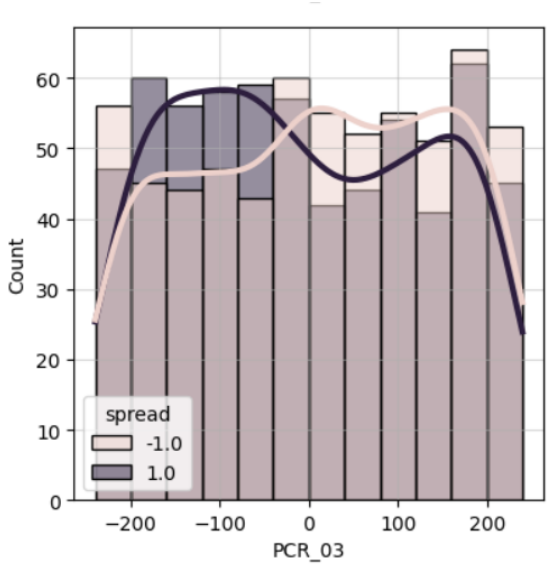
But, the second feature is chisquared meaning most of the points are concentrated close to zero, and the propability to get higher values is smaller as the values get higher , so normalizing the values to the range -1,1 will lead to most of the points to be concentrated around -1, and a very negligible amount will be close to 1

Also , the two features have different ranges, the first is [2,5] , the second's range is [0,infinity], so we would want to normalize them to the same range, but minmax does more harm than good in this case, Z-score normalization would probably be a better option

Part 4: Data Exploration

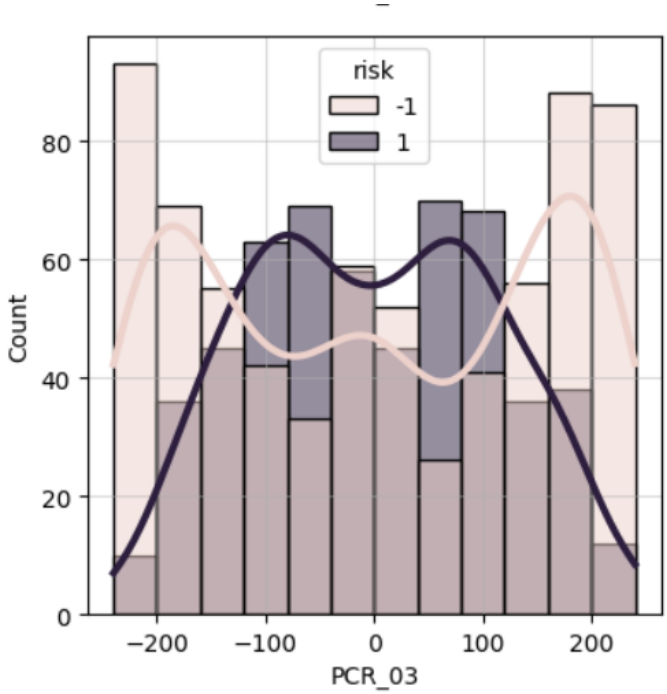
Q14:

PCR_03 seems informative to predict **spread** , as there is good distinction between the two lines, and a clear threshold, for example looking at the graph we can estimate that if PCR_03 is under about -25 , then spread is 1, else if higher than -25 we can predict that it's spread is -1.



Q15:

For the same reasons as Q14, there is clear distinction between the two lines, and we can cut the range into intervals where with good confidence we can predict if risk is 1 or -1, for ex. In the range ~[-120,100] we can predict 1, in other intervals we can predict -1.



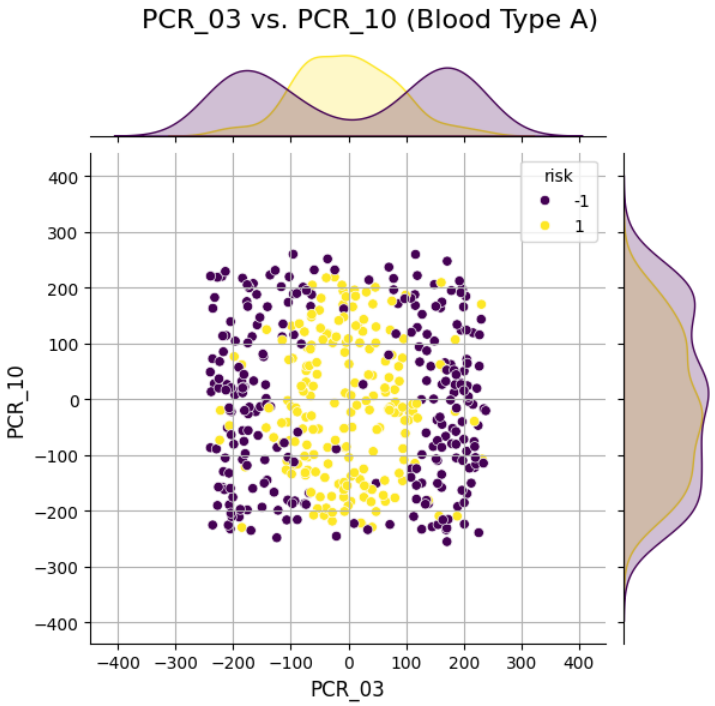
Q16:

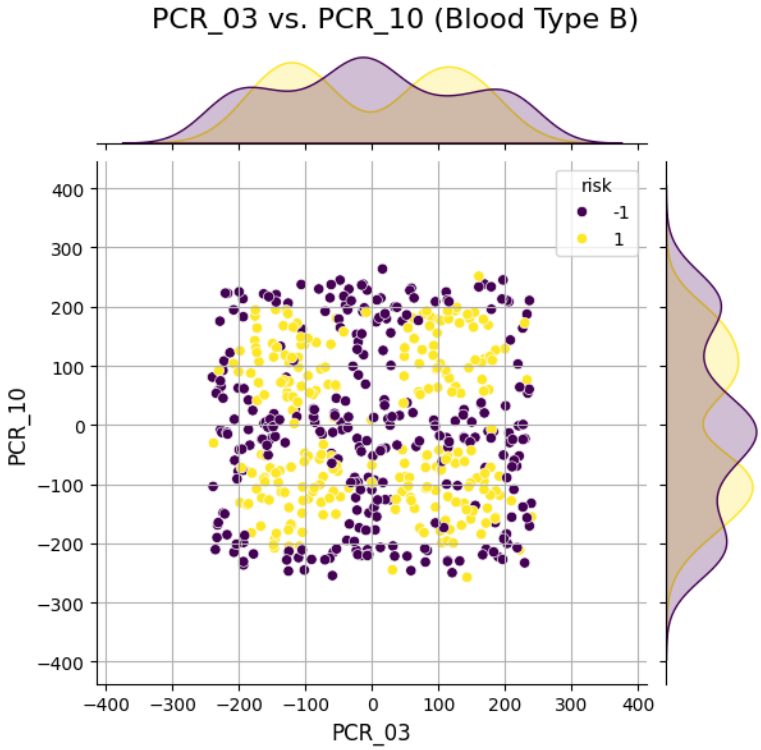
We chose PCR_03 and PCR_10 features because the data shows clear distinction between the groups of points, we can

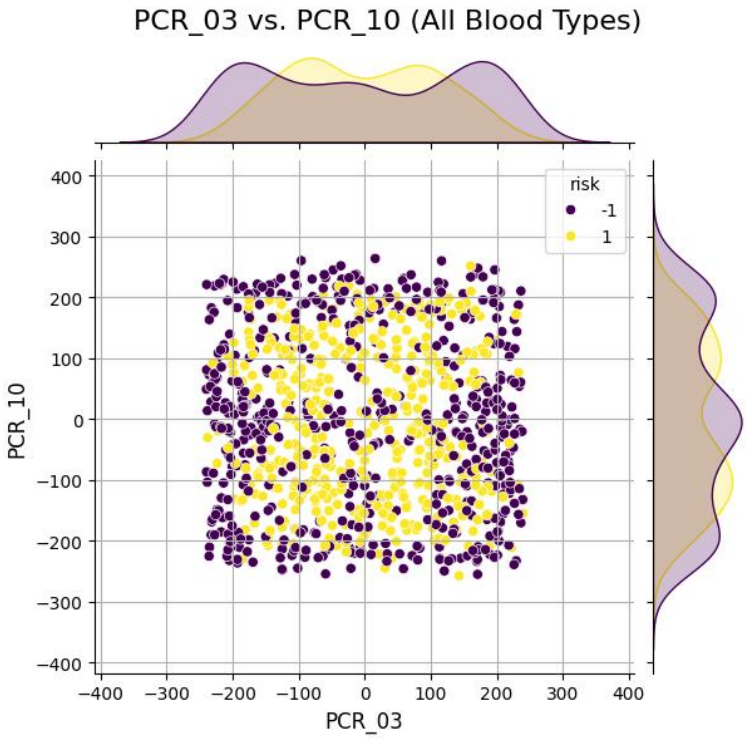
confidently predict given the two features of a patient if he is at risk or not because of the good non-linear separation.

Q17:

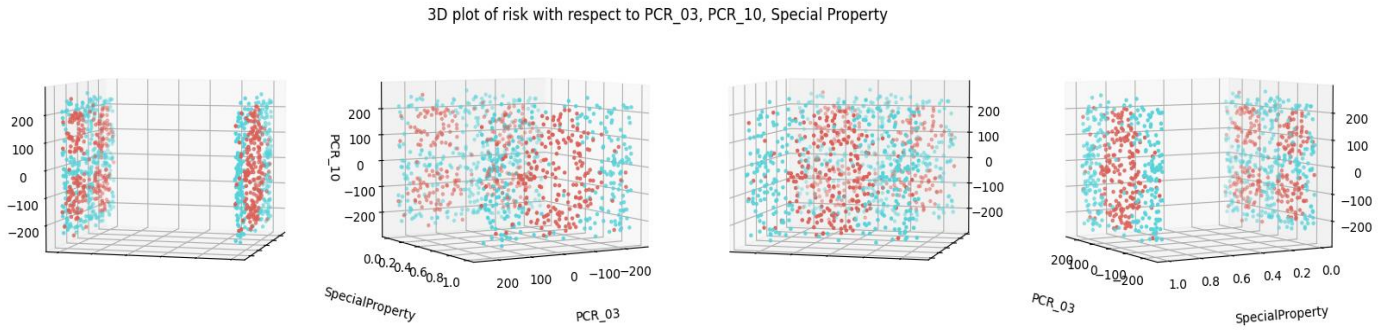
Three join plots:







Q18:



Q19:

Dicision tree of max depth 3:

We could technically use a decision tree of depth 3, but, our prediction wont be the most accurate for the following reason:

The first decision will be based on SpecialProperty to know wether we are group A or B, in the good case, we would be in case group A, because, looking at the graph two lines in the graph- two threshold, would be enough to predict with good

confidence if we are at risk or not since there are three parallel intervals that can be cut using two lines,

In the less fortunate case, we would be in case B, where inside there are 4 rectangles where its more likely to be a risk, however, the two decisions left wouldn't be enough to model the four rectangles, rather, we would need about 4 decisions which should be enough to model a rectangle of 4 sides.

Q20:

Dicision tree of max depth 30:

It could fit the training data **too well**, meaning we might get as a result **overfitting** to the training data that is sensitive to outliers, the 30 splits might be able to reach individual outlier points in the training data and make decision boundaries that are not fit for the test data.

Q21:

A 1-NN model wont fit the data very well, that's because of the very different scales of SpecialProperty against the two PCR's , where the two PCR's ranges are much bigger leading them to dominate the distance calculation, which would make SPecialProperty have much less impact.

Part 5: More Data Normalization

Q22:

Effect of normalization on:	
(Q19)	decision tree's of max depth 3 performance will A nearly not change, since were using minmax scaling, the features values will be downsized by some constant, meaning the 3d plot would look the same but in a smaller range, so if we divide the thresholds that were in the original decision tree by some constant we

	should get similar performance aside from maybe some floating-point rounding errors.
(Q20)	decision tree's of max depth 30 problem of A overfitting won't be fixed by normalization, rather it almost doesn't affect overfitting. However setting a smaller max depth help fix overfitting. normalization would get us close to the same results.
(Q21)	A 1-NN model will fit, thats because the scaling of SpecialProperty and PCR_03 PCR_10 are now more similar meaning they have similar ranges and no feature will be dominant, meaning the distence calculation will be more fair.

Part 6: Data Preparation Pipeline

Q23:

Feature name	**keep**	**New**	**Normalization Method**
Patient_id	V	X	
Age	V	X	
Sex	V	X	
Weight	V	X	
Blood_type	X	X	
Special_Proberty	V	V	
Current_location	V	X	
Num_of_siblings	V	X	

Happiness_score	V	X	
household_income	V	X	
conversations_per_day	V	X	
sugar_levels	V	X	
sport_activity	V	X	
pcr_date	V	X	
PCR_01	V	X	Standard
PCR_02	V	X	Standard
PCR_03	V	X	MinMax
PCR_04	V	X	MinMax
PCR_05	V	X	Standard
PCR_06	V	X	Standard
PCR_07	V	X	Standard
PCR_08	V	X	Standard
PCR_09	V	X	MinMax
PCR_10	V	X	MinMax